

Driver Drowsiness Detection and Alert System

Project Proposal for UCS503
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1 Introduction

This project aims to improve road safety by reducing accidents caused by driver fatigue and micro-sleep episodes. While the underlying concern (fatigue-related crashes) is a broader public-health and transportation issue, the immediate engineering objective is to translate *real-time drowsiness detection* into a measurable, deployable software solution running on commodity hardware (webcam/smartphone camera).

2 Objectives

The primary objective of this project is to develop a lightweight, real-time camera-based system for detecting driver drowsiness and providing timely alerts.

The specific objectives are:

- To detect the driver's face and facial landmarks in real time using a webcam or dashboard-mounted camera.
- To detect prolonged eye closure and yawning using Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR).
- To reduce false alarms using temporal smoothing and per-user calibration.
- To provide an audio/visual alert when sustained signs of drowsiness are detected.
- To evaluate the system using alert latency, detection accuracy, false-alarm rate, and real-time performance.

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3 Problem Statement

Drowsy driving is a significant contributor to road accidents, particularly on highways and during long or night-time drives. Common pain points include:

- **Late detection:** drivers and passengers often fail to notice early signs of fatigue (prolonged blinks, yawning, head nodding) until it is too late.
- **Cost barrier:** existing dedicated drowsiness-detection hardware (e.g., specialized in-cabin sensors) is expensive and not standard in most vehicles, especially in tier-2/3 markets.
- **False alarms:** naive systems using rigid thresholds trigger frequent false positives (e.g., due to glasses, lighting, or eye shape variation), reducing driver trust and adoption.
- **Lack of accessible logging:** there is no easy way for fleet operators or individual users to review drowsiness incidents after a trip.

Why this matters now (contextual relevance): With cameras already available in smart-phones and low-cost dashboard-mounted devices, a software-only or software-plus-commodity-camera solution can meaningfully reduce fatigue-related risk without expensive specialized hardware.

4 Proposed Solution

4.1 Overview

Build a **camera-based** “Drowsiness-to-Alert” system with the following components:

- **Capture module:** continuous video capture from a webcam or dashboard-mounted camera.
- **Face and landmark detection:** locate the driver’s face and extract eye/mouth landmarks per frame.
- **Drowsiness scoring:** compute indicators such as Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to flag prolonged eye closure and yawning.
- **Alert mechanism:** trigger an audio/visual alert when a temporally-smoothed drowsiness score crosses a threshold.
- **Event logging dashboard:** record timestamped drowsiness events (without storing raw video) for later review.

4.2 Core workflow

1. Camera captures a frame; the system detects the driver’s face and facial landmarks.
2. EAR and MAR are computed per frame; a rolling window smooths noisy per-frame values.
3. If eye closure or yawning persists beyond a calibrated duration, a drowsiness state is flagged.
4. An alert (sound/vibration/visual) is triggered, and the event is logged with a timestamp and severity level.

4.3 Operational constraints for an educational, real-time setting

- Keep the pipeline lightweight enough to run in real time (target ≥ 15 FPS) on low-to-mid range hardware (e.g., laptop webcam, Raspberry Pi, or a mid-range smartphone).
- Avoid dependence on training novel deep-learning models from scratch; use pre-trained facial landmark detectors and well-established heuristics.
- Design for deployability as a standalone desktop/mobile app with an optional lightweight backend for logging and fleet-level aggregation.

5 Solution Approach

This is an engineering course project, so we focus on integrating mature computer-vision components into a reliable, real-time, measurable pipeline rather than researching new detection algorithms.

5.1 Drowsiness scoring

Detection will be heuristic-based, built on established, well-documented techniques:

- Use a pre-trained facial landmark model (e.g., dlib’s 68-point model or MediaPipe Face Mesh) rather than training a new detector.
- Compute EAR (eye aspect ratio) and MAR (mouth aspect ratio) using standard published formulas, with calibrated per-user thresholds.
- Apply temporal smoothing (e.g., a sliding window / consecutive-frame count) to reduce false positives from single-frame noise, and surface borderline cases as “low-confidence” alerts rather than silently suppressing them.

5.2 System architecture

A layered architecture with an edge-first design:

- **Edge/client layer:** camera capture, face/landmark detection, scoring, and alerting run locally to minimize latency and avoid transmitting raw video.
- **Backend API (optional):** receives only lightweight event metadata (timestamp, severity, session ID) for logging and analytics; no raw frames are uploaded by default.
- **Database:** stores users, sessions, drowsiness events, and calibration profiles.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically run unit tests on the scoring logic (EAR/MAR computation, thresholding, smoothing) on every change.
- Run pipeline tests against a fixed set of labeled sample videos to catch regressions in detection behavior.
- Produce repeatable deployment artifacts for the desktop/mobile client and backend to staging.

6 Evaluation Criterion

We define success using measurable metrics tied to detection accuracy and real-time responsiveness.

6.1 Primary evaluation metric

Alert latency: time between the onset of a drowsiness event (e.g., sustained eye closure) in the input video and the system raising an alert.

- **Target:** median alert latency ≤ 1 second on labeled test clips.
- **Attribution:** measured by comparing ground-truth event onset timestamps (from labeled/annotated test videos) against logged alert timestamps.

6.2 Secondary evaluation metrics

- **Detection accuracy:** sensitivity (recall) and specificity of drowsiness classification against a labeled validation set (e.g., a subset of an existing public drowsiness dataset or self-recorded, consented clips).
- **False alarm rate:** number of false-positive alerts per hour of normal (alert) driving footage.
- **Real-time performance:** sustained frames-per-second on target hardware, and CPU/memory usage.
- **User-perceived reliability:** a simple 1–5 feedback question on alert timeliness and annoyance level from pilot users.

6.3 Pilot validation plan

- Collect or use an existing small set of labeled driving/alertness video clips (with consent, or a public dataset) covering both alert and simulated-drowsy states.
- Run 5–10 pilot users through short supervised sessions (e.g., a driving simulator or seated webcam test) to measure real-world alert latency and false-alarm rate.

7 Timeline

Weeks	Task	Deliverable
1–2	Literature review, requirement analysis, and project setup	Requirements specification and technology stack
3–4	Camera capture and facial landmark detection	Working face and landmark detection module
5–6	EAR-based eye-closure detection and threshold calibration	Eye-closure detection prototype
7–8	MAR-based yawning detection and combined drowsiness scoring	Drowsiness scoring module
9–10	Alert mechanism and event logging integration	Working end-to-end detection and alert system
11	Testing, CI/CD integration, and performance evaluation	Test results and performance report
12	Buffer week, refinement, documentation, and final testing	Final prototype and project documentation

8 Scalability

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth from single-driver monitoring to fleet-level analytics by:
 - keeping heavy processing on the edge device so backend load only scales with event metadata, not video volume,
 - using stateless, indexed APIs for querying historical events across many drivers/vehicles,
 - allowing per-user calibration profiles to be stored and retrieved independently.
2. **Operationally deployable (practically):** The system should run on common hardware and hosting:
 - lightweight client for laptops, Raspberry Pi-class devices, or smartphones,
 - managed database and containerized backend for the optional logging service,
 - CI/CD pipeline for staging/production releases of both client and backend.

9 Technology Stack

A mature set of “engines” (tooling/libraries) is available to implement this without inventing new computer-vision research:

- Computer-vision library for frame capture and preprocessing (OpenCV).
- Pre-trained facial landmark detector (dlib 68-point model or MediaPipe Face Mesh).
- Audio/notification library for triggering alerts on the client device.
- Lightweight web/mobile framework for the client UI and optional dashboard.
- Managed relational or document database for event logging.

- Test framework for unit/integration tests on scoring logic.
- CI runner and staging deployment target.

We will use these mature components to focus on pipeline integration, calibration, correctness, and measurement rather than building detection models from scratch.

10 Project scope and deliverables

10.1 Initial deliverable

- Webcam capture + facial landmark detection integration
- EAR-based eye-closure detection with basic thresholding and temporal smoothing
- Local audio/visual alert on sustained eye closure
- Basic event logging with timestamps for latency measurement
- CI: build + unit tests for scoring logic + linting
- CD to staging for the client application

10.2 Subsequent deliverables

- MAR-based yawning detection and combined drowsiness scoring
- Per-user calibration flow to reduce false alarms
- Optional backend for event logging and a simple analytics dashboard
- Low-light/IR-camera considerations for night-time driving scenarios

11 Risks and mitigations

- **Privacy of video data:** process video entirely on-device where possible; never store or transmit raw frames by default, only derived event metadata.
- **False positives/negatives:** mitigate via per-user calibration, temporal smoothing, and clearly logging low-confidence events instead of silently discarding or over-alerting.
- **Hardware/lighting variability:** test across multiple lighting conditions and camera qualities early; document known limitations (e.g., sunglasses, extreme low light).
- **Driver distraction from alerts:** keep alert design (sound/vibration pattern) simple and non-startling, validated informally with pilot users.
- **Operational issues in pilot:** use staging and recorded test videos early; ensure CI/CD produces repeatable, testable builds before live testing.

12 Summary

This project proposes a deployable, camera-based drowsiness prediction and alerting system to reduce fatigue-related driving risk. It meets the course criteria by emphasizing time-to-value through rapid iterations built on pre-trained, mature computer-vision components, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially alert latency and detection accuracy). It remains focused on engineering integration, real-time reliability, and scalability readiness rather than novel algorithmic research.